

FasterCap Output

Output text

FasterCap, while running, provides many informations about what it is doing, plus the result of the calculations.

Here below you can see a copy of FasterCap output text for the capacitance calculation of the two cubes problem, as shown in the [Input File Structure](#) section.

In the next pages you can find more information about each section of the output text

- [Output header](#)
- [Output Iteration information](#)
- [Output tail](#)

```
Running FasterCap version 5.0.2
Copyright 2013 FastFieldSolvers S.r.l. http://www.fastfieldsolvers.com, All Rights reserved
Starting capacitance extraction with the following parameters:
Input file: C:\Documents and Settings\All Users\Documents\FastFieldSolvers\FasterCap
\cubes.lst
Auto calculation with max error: 0.01
Remark: Auto option overrides all other Manual settings

Solution scheme (-g): Collocation, GMRES tolerance (-t): 0.005
Out-of-core free memory to link memory condition (-f): 5
Potential interaction coefficient to mesh refinement ratio (-d): 1
Mesh curvature (-mc): 3
Number of input panels to solver engine: 24

Iteration number #0 *****

*****
Refining the geometry..
Refinement completed
Mesh refinement (-m): 1e+032
*****
Computing the links..
Number of panels after refinement: 24
Number of links to be computed: 46
Done computing links
*****
Precond Type(s) (-p): Jacobi
GMRES Iteration: 0 1
GMRES Iteration: 0 1
Capacitance matrix is:
Dimension 2 x 2
g1_mycube 6.49181e-011 -2.07808e-011
g2_mycube -1.76613e-011 6.49235e-011

Solve statistics:
Number of input panels: 12 of which 12 conductors and 0 dielectric
Number of input panels to solver engine: 24
Number of panels after refinement: 24
Number of potential estimates: 23
Number of links: 70 (uncompressed 576, compression ratio is 87.8%)
Max recursion level: 5
Max Mesh relative refinement value: 0.0533222
Iteration time: 0.069000s (0 days, 0 hours, 0 mins, 0 s)
Iteration allocated memory: 12330 kilobytes

Iteration number #1 *****

*****
Increasing the geometric refinement..
Refinement completed
Mesh refinement (-m): 0.0377045
*****
Computing the links..
Number of panels after refinement: 96
Number of links to be computed: 4628
Done computing links
*****
Precond Type(s) (-p): Jacobi
```

```

GMRES Iteration: 0 1 2
GMRES Iteration: 0 1 2
Capacitance matrix is:
Dimension 2 x 2
g1_mycube 8.02788e-011 -2.59788e-011
g2_mycube -2.59743e-011 8.02738e-011

```

Weighted Frobenius norm of the difference between capacitance (auto option): 0.199681

Solve statistics:

```

Number of input panels: 12 of which 12 conductors and 0 dielectric
Number of input panels to solver engine: 24
Number of panels after refinement: 96
Number of potential estimates: 2314
Number of links: 4724 (uncompressed 9216, compression ratio is 48.7%)
Max recursion level: 13
Max Mesh relative refinement value: 0.0373801
Iteration time: 0.035000s (0 days, 0 hours, 0 mins, 0 s)
Iteration allocated memory: 12360 kilobytes

```

Iteration number #2 *****

```

*****
Increasing the geometric refinement..
Refinement completed
Mesh refinement (-m): 0.0264317
*****
Computing the links..
Number of panels after refinement: 192
Number of links to be computed: 11048
Done computing links
*****
Precond Type(s) (-p): Jacobi
GMRES Iteration: 0 1 2 3
GMRES Iteration: 0 1 2 3
Capacitance matrix is:
Dimension 2 x 2
g1_mycube 8.1187e-011 -2.64358e-011
g2_mycube -2.64368e-011 8.11959e-011

```

Weighted Frobenius norm of the difference between capacitance (auto option): 0.011994

Solve statistics:

```

Number of input panels: 12 of which 12 conductors and 0 dielectric
Number of input panels to solver engine: 24
Number of panels after refinement: 192
Number of potential estimates: 5524
Number of links: 11240 (uncompressed 36864, compression ratio is 69.5%)
Max recursion level: 14
Max Mesh relative refinement value: 0.0264134
Iteration time: 0.057000s (0 days, 0 hours, 0 mins, 0 s)
Iteration allocated memory: 12402 kilobytes

```

Iteration number #3 *****

```

*****
Increasing the geometric refinement..
Refinement completed
Mesh refinement (-m): 0.0186771
*****
Computing the links..
Number of panels after refinement: 288
Number of links to be computed: 21936
Done computing links
*****
Precond Type(s) (-p): Jacobi
GMRES Iteration: 0 1 2 3
GMRES Iteration: 0 1 2 3
Capacitance matrix is:
Dimension 2 x 2
g1_mycube 8.1964e-011 -2.6891e-011
g2_mycube -2.68863e-011 8.19668e-011

```

Weighted Frobenius norm of the difference between capacitance (auto option): 0.0103921

Solve statistics:

```

Number of input panels: 12 of which 12 conductors and 0 dielectric
Number of input panels to solver engine: 24
Number of panels after refinement: 288
Number of potential estimates: 10968

```

Number of links: 22224 (uncompressed 82944, compression ratio is 73.2%)
 Max recursion level: 16
 Max Mesh relative refinement value: 0.0186339
 Iteration time: 0.058000s (0 days, 0 hours, 0 mins, 0 s)
 Iteration allocated memory: 12444 kilobytes

Iteration number #4 *****

Increasing the geometric refinement..

Refinement completed

Mesh refinement (-m): 0.0131762

Computing the links..

Number of panels after refinement: 480

Number of links to be computed: 45824

Done computing links

Precond Type(s) (-p): Jacobi

GMRES Iteration: 0 1 2 3 4

GMRES Iteration: 0 1 2 3 4

Capacitance matrix is:

Dimension 2 x 2

g1_mycube 8.26757e-011 -2.73819e-011

g2_mycube -2.73807e-011 8.26804e-011

Weighted Frobenius norm of the difference between capacitance (auto option): 0.00994789

Solve statistics:

Number of input panels: 12 of which 12 conductors and 0 dielectric

Number of input panels to solver engine: 24

Number of panels after refinement: 480

Number of potential estimates: 22912

Number of links: 46304 (uncompressed 230400, compression ratio is 79.9%)

Max recursion level: 18

Max Mesh relative refinement value: 0.0131762

Iteration time: 0.092000s (0 days, 0 hours, 0 mins, 0 s)

Iteration allocated memory: 12530 kilobytes

Total allocated memory: 12530 kilobytes

Total time: 0.311000s (0 days, 0 hours, 0 mins, 0 s)